DSAIT4335 Recommender Systems

Grading protocol

Knock-out criteria (you need to meet the following to qualify for grading):

- All necessary files are submitted (report in PDF and source codes).
- The report is structured according to the guideline introduced in project definition.
- Grammatical and spelling errors, as well as figure and table formats, do not severely hinder readability.

The final grade is calculated as a weighted sum of all rubric item scores, with performance levels ranging from 1 point for Insufficient to 4 points for Strong, and rescaled to a range from 1 to 10.

	Insufficient 1 point	**Developing** 2 points	**Proficient** 3 points	**Strong** 4 points
B. Build the models				
B1. Select and tune individual recommenders **(10%)**	Model selection or tuning is absent or inadequately justified. Configurations are inappropriate, or the process and decisions cannot be established from the explanation.	Models are selected and configured, but choices are supported mainly by generic reasons. Tuning is incomplete or its procedure and outcomes are insufficiently explained.	Model selection and configuration are appropriate for the task and data. Tuning is conducted systematically, with choices justified and the process explained sufficiently to reproduce it.	Model selection reflects a reasoned assessment of the suitability, strengths, and limitations of alternative methods. Tuning decisions are informed by validation evidence, and relevant constraints and uncertainties are considered.
B2. Build hybrid recommenders **(10%)**	Required hybrids are absent or fundamentally incorrect. Component combination or coefficient learning is unjustified or inadequately explained.	Hybrids are operational, but combination or tuning procedures have weaknesses. Reasons for choosing components and combining them are partial, and important details are unclear.	Required hybrids are correctly constructed and tuned. Component selection, combination mechanisms, and coefficient learning are justified. The process is explained sufficiently to reproduce it.	Design choices explicitly address how components are expected to complement one another and how their outputs interact. Combination and tuning decisions are supported by validation evidence, with relevant limitations explained.
B3. Build reranking interventions **(10%)**	Required interventions are absent or fundamentally incorrect. Their objectives or integration with the recommenders are unclear.	Interventions are operational, but implementation, integration, or parameter selection has weaknesses. Their selection is only partly connected to the intended objectives.	Required interventions are correctly implemented and integrated. Selection and parameter choices are justified against explicit objectives, with appropriate tuning where applicable. The process is explained sufficiently to reproduce it.	Intervention design and placement reflect a reasoned assessment of objectives, assumptions, and constraints. Parameter decisions are supported by appropriate evidence, and the rationale explains the intended balance between competing objectives.
U. Understand the models				
U1. Understand individual recommenders **(15%)**	Little or no analysis of how the models behave. Conclusions are absent, unsupported, or limited to restating performance results.	Some relevant analysis is provided, but explanations of model behavior are generic or incomplete. Connections between model characteristics, data properties, and observed results are weak.	Analysis explains relevant behavioral patterns in the individual models and connects them to their underlying mechanisms and the characteristics of the data. Conclusions are supported by appropriate evidence.	Analysis develops deeper insights into why models behave differently, identifies relevant strengths, limitations, or failure patterns, and uses evidence to examine explanations or generate new hypotheses.
U2. Understand hybrid recommenders	Little or no analysis of how the hybrid behaves or how its components contribute.	Some component-level analysis is provided, but the contributions and interactions of	Analysis explains how individual components contribute to hybrid behavior and how their	Analysis reveals deeper insights into component complementarity, redundancy, or interactions, and identifies

(15%)	Conclusions about the hybrid are unsupported or limited to aggregate performance comparisons.	individual recommenders are only partly explained. Conclusions about complementarity or redundancy are weakly supported.	strengths, limitations, and interactions relate to the observed results. Conclusions are supported by coefficient analysis or other relevant evidence.	conditions under which particular components contribute more or less. Insights are used to derive well-motivated hypotheses about how the hybrid could be improved.
U3. Understand reranking interventions (10%)	Little or no analysis of how reranking changes recommendations or why the changes occur. Conclusions are unsupported or limited to reporting before-and-after results.	Some changes in recommendation behavior are identified, but explanations of the effects of the intervention are incomplete or weakly connected to its mechanism and objectives.	Analysis explains how and why the intervention changes recommendation behavior in relation to its objectives. Conclusions are supported by evidence about the resulting recommendations and relevant behavioral patterns.	Analysis reveals deeper insights into the mechanisms, conditions, or limitations that determine the intervention's effects. Findings are used to identify opportunities for improvement or develop testable hypotheses about alternative intervention designs.
E. Evaluate recommendation behavior and performance				
E1. Evaluate individual recommenders (10%)	Evaluation is missing, incorrect, or inadequate to support conclusions. Accuracy is assessed incompletely, or beyond-accuracy dimensions are omitted. Comparisons with baselines are missing or uninterpreted.	Accuracy and some beyond-accuracy dimensions are evaluated, but metric selection, baselines, or interpretation are incomplete. Results are mainly descriptive, and differences across user/item groups are insufficiently examined.	Accuracy and relevant beyond-accuracy dimensions are assessed using appropriate metrics and baselines. Results are interpreted in relation to the objectives of the study, including differences across relevant user/item groups. Conclusions are supported by the evidence.	Evaluation provides a nuanced account of where models succeed and fail across quality dimensions and user/item groups. Differences between accuracy and beyond-accuracy performance are explained, and the limitations of the evaluation are considered.
E2. Evaluate hybrid recommenders (10%)	Evaluation does not establish whether hybridization improves upon individual recommenders or relevant baselines. Accuracy and beyond-accuracy performance are inadequately assessed, and conclusions are unsupported.	Hybrid performance is compared with individual recommenders, but evaluation across quality dimensions is incomplete or interpretation does not establish when hybridization helps or fails. Differences across user/item groups are only superficially considered.	The hybrid is evaluated against its individual components and relevant baselines using appropriate accuracy and beyond-accuracy metrics. Results establish the benefits and limitations of hybridization and examine how performance varies across relevant user/item groups.	Evaluation explains when and why hybridization provides benefits or introduces disadvantages across quality dimensions and user/item groups. It assesses whether gains in one dimension are accompanied by losses in another and uses the findings to develop a reasoned assessment of the hybrid's added value.
E3. Evaluate reranking interventions and societal implications (10%)	Reranking effects are not adequately evaluated against their intended objectives. Societal or beyond-accuracy considerations are missing, or conclusions are based on isolated before-and-after metric comparisons.	Some effects of reranking on accuracy and beyond-accuracy metrics are assessed, but the evaluation does not adequately explain the trade-offs, affected user/item groups, or implications for the intended objectives.	Evaluation assesses reranking effects on accuracy and relevant beyond-accuracy dimensions. Trade-offs are explained using evidence, and effects on relevant user/item groups are examined.	Evaluation critically examines trade-offs between accuracy, beyond-accuracy objectives, and societal implications. It explains how effects vary across intervention settings and affected groups, considers alternative intervention or pipeline designs, and provides a reasoned judgment of the benefits, limitations, and distribution of outcomes.

* **Peer evaluation** affects the grade for group evaluation part; in case of negative feedback from peers about the contribution and performance in group, the final grade will be adjusted.